

Dublin Business School

B9DA109 — Machine Learning and Pattern Recognition

Predicting Student Exam Performance Using Supervised Machine Learning

Decision Tree, Random Forest and SVR Regression Models

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1. Problem Definition, Dataset and Algorithm Selection

1.1 Problem

This project asks a simple but practically useful question: can we predict how well a student will score in their end-of-year exam based on measurable factors like their study habits, attendance, access to resources, and home environment? Framing this as a supervised regression problem is the natural choice because Exam_Score is a continuous numeric variable. A working model would allow schools and teachers to identify students who appear on track for a poor result early enough to offer targeted support.

1.2 Dataset

The dataset used is the Student Performance Factors dataset (Kaggle). It contains 6,607 student records with 20 original columns. After removing five features on domain grounds (see Section 3), 14 independent variables remain alongside the target Exam_Score. Two columns had small numbers of missing values: Teacher_Quality (78 missing) and Parental_Education_Level (90 missing), both filled with their respective column mode before any modelling.

1.3 Algorithm Selection

Three supervised regression algorithms were selected to represent a deliberate range of model complexity:

- Decision Tree Regressor: the interpretable baseline. A single tree with a controlled depth is fast, readable, and gives a performance floor to beat.
- Random Forest Regressor: a bagging ensemble. It builds many trees independently and averages their predictions, which reduces the variance problem that a single tree suffers from at greater depths.
- Support Vector Regressor: a kernel-based method. SVR maps features into a higher-dimensional space via a kernel function, allowing it to model non-linear relationships without explicitly engineering interaction terms. Its epsilon-insensitive loss also makes it robust to individual students with unusual profiles.

Together, the three span the range from a transparent baseline to a complex kernel model, giving a meaningful basis for comparison.

2. Data Preparation

2.1 Feature Removal

Five columns were removed before any analysis: Gender, School_Type, Physical_Activity, Sleep_Hours, and Distance_from_Home. These were judged unlikely to carry a consistent directional signal in predicting Exam_Score — Gender and School_Type have no meaningful ordinal relationship with exam outcomes; the remaining three are peripheral to the core academic preparation factors the dataset is designed to capture. Removing them reduces noise and keeps the model focused. The remaining 14 features all relate to study behaviour, motivation, support systems, or prior academic history.

2.2 Data Cleaning/Missing Value Imputation

Teacher_Quality had 78 missing entries (approximately 1.2% of rows), and Parental_Education_Level had 90 (approximately 1.4%). Both were filled with the mode of their respective column — the most frequently occurring category. Mode imputation was chosen because these are ordinal categorical variables, and mean imputation would produce fractional values that correspond to no real category. Filling with the mode is conservative: it does not

introduce a new value that wasn't already present in the data, and because the missing proportion is small, it has a minimal effect on the overall distribution.

2.3 Data Encoding

Ten of the 14 remaining features were categorical strings. Each was mapped to an ordered integer using pandas .map(). Ordinal encoding was chosen over one-hot encoding because every column has a natural order that should be preserved — converting Low/Medium/High to 0/1/2 keeps the gradient intact, whereas one-hot encoding would treat each level as completely independent. The mappings used were:

- Parental_Involvement, Access_to_Resources, Motivation_Level, Teacher_Quality, Family_Income: Low=0, Medium=1, High=2
- Extracurricular_Activities, Internet_Access, Learning_Disabilities: No=0, Yes=1
- Peer_Influence: Neutral=0, Negative=1, Positive=2
- Parental_Education_Level: High School=0, College=1, Postgraduate=2

2.4 Data Scaling

StandardScaler was applied to all 14 independent variables after encoding, transforming each to zero mean and unit variance. This is a hard requirement for SVR, which computes distances between data points — without scaling, features measured in large absolute values (like Hours_Studied: 1–44) would dominate features like Internet_Access (0/1). Scaling also aids the other models during grid search by keeping parameter ranges comparable across features.

2.5 Train/Test Split

A 90/10 split was used with random_state=100, producing 5,946 training rows and 661 test rows. A 90/10 ratio was chosen deliberately given the moderate dataset size of 6,607 rows — maximising the training data is more valuable than a larger test set when the total sample is not especially large.

3. Feature Selection

Feature selection was applied at the start of the pipeline using a domain-based filter method. The five removed columns (Gender, School_Type, Physical_Activity, Sleep_Hours, Distance_from_Home) were excluded because they do not represent factors that a student, teacher, or institution can directly act on, and because the existing literature on academic performance does not consistently identify them as strong predictors of exam scores when controlling for study and support factors.

The 14 retained features all belong to categories that the educational psychology literature associates with academic achievement: prior performance (Previous_Scores), study engagement (Hours_Studied, Attendance, Tutoring_Sessions), motivation and support (Motivation_Level, Parental_Involvement, Family_Income, Internet_Access, Access_to_Resources, Teacher_Quality), and peer and disability factors (Peer_Influence, Learning_Disabilities, Parental_Education_Level, Extracurricular_Activities).

No statistical filter (e.g. mutual information or correlation threshold) was applied after this initial selection. Given that the best model achieved a test R^2 of 84.49%, the 14-feature set is clearly doing effective predictive work.

4. Model Development and Evaluation

4.1 Evaluation Metric

R^2 (coefficient of determination) was used throughout as the primary metric. For regression on Exam_Score, R^2 reports what fraction of the variance in exam scores the model accounts for. An R^2 of 0.84 means the model explains 84% of the variation in outcomes — intuitive and comparable across models without needing to interpret units.

4.2 Decision Tree Regressor

A baseline Decision Tree with `max_depth=3` was built and evaluated on the 661-row test set, returning $R^2 = 54.68\%$. At depth 3 the tree makes only three levels of splits, which limits it to capturing only the broadest patterns in the data. `GridSearchCV` was then run over `max_depth` values 1 through 99 with 5-fold cross-validation, identifying `max_depth=5` as optimal with a cross-validated R^2 of 50.30%. The CV score being lower than the test score suggests the fixed test split happened to be a slightly easier sample.

4.3 Random Forest Regressor

A Random Forest baseline with 150 trees was trained and tested, returning $R^2 = 78.86\%$ — a 24-percentage-point improvement over the Decision Tree baseline, demonstrating the value of the ensemble approach. `GridSearchCV` over `{50, 100, 150, 200, 250}` estimators with 5-fold CV found 200 trees to be optimal, giving a cross-validated R^2 of 63.72%. Again the test score exceeds the CV score, consistent with the pattern seen in the Decision Tree.

4.4 Support Vector Regressor

A baseline SVR with a polynomial kernel was trained on the 90% training set and evaluated on the test set, returning $R^2 = 84.49\%$ — the highest of any model tested. `GridSearchCV` was then run across all combinations of kernels `{linear, rbf, poly, sigmoid}` and regularisation values `C = {0.01, 0.1, 1, 10, 100}` with 5-fold cross-validation. The optimal configuration was `C=100` with a linear kernel, yielding a cross-validated R^2 of 71.17%. The linear kernel outperforming poly and rbf in cross-validation is an interesting finding: it suggests that once the regularisation strength is tuned correctly, the relationship between the 14 features and Exam_Score is reasonably well captured by a linear decision boundary in the scaled feature space.

4.5 Results Summary

Model & Configuration	R^2 Score	Evaluation Type
Decision Tree (baseline, <code>max_depth=3</code>)	54.68%	Test R^2
Decision Tree (optimised, <code>max_depth=5</code> , CV)	50.30%	Cross-validated R^2
Random Forest (baseline, <code>n_estimators=150</code>)	78.86%	Test R^2
Random Forest (optimised, <code>n_estimators=200</code>)	63.72%	Cross-validated R^2
SVR (baseline, <code>kernel='poly'</code>) ★ Best	84.49%	Test R^2
SVR (optimised, <code>C=100</code> , linear kernel)	71.17%	Cross-validated R^2

5. Optimisation

All three models were subjected to hyperparameter optimisation via `GridSearchCV` with 5-fold cross-validation, using R^2 as the scoring metric throughout.

- Decision Tree: `max_depth` searched exhaustively over 1–99. The grid search settled on depth 5, a slightly deeper tree than the baseline depth of 3. The cross-validated score (50.30%) falling below the single-split test score (54.68%) indicates the baseline test evaluation was optimistic — the 661-row test set appears to be somewhat easier than the full dataset on average.
- Random Forest: `n_estimators` searched over {50, 100, 150, 200, 250}. Optimal at 200 trees. The marginal gain from 150 to 200 trees is modest and follows the expected variance-reduction curve for bagging ensembles. Further tree additions beyond 200 produced no benefit in the tested range.
- SVR: 20 parameter combinations (4 kernels $\times$ 5 C values) evaluated across 5 folds (100 total model fits). Optimal: `C=100`, linear kernel. A high C value means the model allows less regularisation and fits the training data more closely — appropriate here where the relationship between study/support features and exam score appears to be strong and consistent rather than noisy.

6. Model Comparison and Recommendation

6.1 Comparison

Ranking the three models on their primary test-set R^2 :

- SVR (poly kernel, baseline): 84.49% — highest test performance
- Random Forest (150 trees, baseline): 78.86%
- Decision Tree (depth 3, baseline): 54.68%

A consistent pattern runs through all three: the baseline test scores are higher than the cross-validated optimised scores. This is not a sign that the models are poorly specified — it reflects the fact that a single 90/10 split can produce a test set that happens to be easier than average. Cross-validation gives a more honest estimate by averaging performance across five held-out folds. Even so, SVR leads on both measures: 84.49% test R^2 and 71.17% cross-validated R^2 .

6.2 Recommended Model: Support Vector Regressor

SVR is selected as the best model for this problem, for the following reasons:

- Performance: 84.49% test R^2 is the highest of any model at any configuration. It explains the majority of variance in `Exam_Score` from the 14 available features.
- Cross-validation consistency: 71.17% cross-validated R^2 is also the highest, confirming the lead is not a test-set artefact.
- Robustness: the epsilon-insensitive loss function means individual students with atypical exam scores do not disproportionately distort the model, making it more reliable in a real school setting where some student records may be unusual.
- Interpretability of the tuned result: the linear kernel winning in `GridSearchCV` tells us something substantive — the core relationships in this dataset (study hours $\rightarrow$ exam score; attendance $\rightarrow$ exam score) are approximately linear once features are scaled, which gives some degree of interpretability to the model's predictions.

6.3 Limitations and Future Work

- The gap between test and CV scores for all three models suggests a moderately lucky train/test split. A larger dataset or a repeated k-fold evaluation would give more stable estimates.
- Mode imputation is conservative but could be refined — for instance, using a KNN imputer to fill `Teacher_Quality` based on similar school profiles.

- Ordinal encoding assumes equal-step intervals between Low, Medium, and High categories, which may not hold in practice. Target encoding or embedding approaches could be explored.
- Future work: adding RMSE and MAE alongside R^2 would give a clearer picture of prediction error in actual exam-score units, making the results more communicable to non-technical educational stakeholders.

Conclusion

This project set out to predict student exam scores from 14 academic and contextual features using three supervised regression models. Across the full pipeline, from domain-based feature selection and mode imputation through to ordinal encoding, scaling, and hyperparameter tuning, the Support Vector Regressor emerged as the strongest performer, achieving a test R^2 of 84.49% with a polynomial kernel and a cross-validated R^2 of 71.17% after GridSearchCV tuned it to a linear kernel with $C=100$.

The results tell a clear story about model complexity and this particular dataset. The Decision Tree, even after optimisation to $\text{max_depth}=5$, topped out at 50.30% cross-validated R^2 , it simply lacks the capacity to capture the interactions among study hours, attendance, motivation, and support factors that drive exam performance. The Random Forest improved substantially on this, achieving a cross-validated R^2 of 63.72%, confirming that an ensemble approach adds real value. The SVR outperformed both, and the fact that GridSearchCV selected a linear kernel over poly and rbf is itself informative: once features are properly scaled, the relationships between the predictors and Exam_Score are reasonably linear, which gives some confidence that the model is capturing genuine patterns rather than over-fitting to noise.

One finding worth acknowledging honestly is that all three baseline test scores exceeded their cross-validated equivalents. This gap points to the 90/10 split producing a test set that was slightly easier than the full dataset on average, cross-validation corrects for this by rotating the held-out fold across the entire dataset. The SVR's lead holds on both measures, but future work should use repeated k-fold evaluation to produce more stable performance estimates.

In practice, a model of this accuracy could be deployed as an early-warning tool within a school system. Given the 14 input features, all of which are either already collected or straightforwardly measurable, a teacher or pastoral advisor could receive a predicted exam score for each student mid-term and flag those falling below a threshold for additional tutoring or parental engagement. Feature importance analysis on the Random Forest, or SHAP values applied to the SVR, could then explain which factors are driving a particular student's predicted outcome, making the intervention targeted rather than generic.

References

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